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PES UNIVERSITY	**PES University, Bengaluru** **(Established under Karnataka Act No. 16 of 2013)**	**UE20CS931**

February 2025: END SEMESTER ASSESSMENT (ESA)
M TECH DATA SCIENCE AND ARTIFICIAL INTELLIGENCE_ SEMESTER II

UE20CS931- MACHINE LEARNING - II

Time: 3 Hrs	Answer All Questions	Max Marks: 100

Instructions

1. Answer all the questions.
2. Section A should be handwritten in the answer script provided.
3. Section B and C are coding questions to be answered in the system and uploaded.
4. Periodically save a Jupyter notebook, after successful run of each cell.
5. Write appropriate inferences.

Section A (20 marks)			
1	a)	Define precision, recall, and F1-score in the context of a classification problem. Explain their significance and how they are used to evaluate a model's performance.	4
	b)	Explain briefly about Gaussian naive Bayes classification algorithm. State its assumptions.	4
	c)	Explain briefly about stacking and voting classifier.	4
	d)	Explain the concept of bagging in machine learning. How does it improve the performance of a classification model? Explain the training and testing process of random forest algorithm.	5
	e)	What is cross-validation in machine learning?	3
Section B (40 Marks)			
2		This dataset contains information about employees in a company, including their educational backgrounds, work history, demographics, and employment-related factors. It has been anonymized to protect privacy while still providing valuable insights into the workforce. (Refer jupyter notebook for detail of the dataset)	
	(a)	Read the dataset and print the following (8 marks) ● Describe the structure of a given dataset in terms of its shape, the number of numerical and categorical variables, and provide descriptive statistics for both types of variables. Additionally, mention any key observations based on the descriptive statistics. (5 mark) ● Summarize the categorical variables in a dataset by identifying the number of unique categories and the proportion of observations in each category. Highlight any noticeable patterns or anomalies. (3 mark)	8
	(b)	Examine missing values, outliers by plotting . Examine is the Target variable evenly balanced.	8
	(c)	Perform appropriate encoding on the categorical attributes.	6
	(d)	Perform necessary actions to ‘fix’ defects in the data ● Impute the missing values using central values (Mean, median, mode)	8

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		● Detect the outliers and decide if any treatment is required. ● Detect the unnecessary features and remove.	
	(e)	Examine the correlation and summarize the relationship between variables. Use appropriate plots to justify the same.	6
	(f)	Split dataset into train and test (70:30)	4
Section C (40 marks)			
3	(a)	Fit a Logistic Regression model and compare the results with respect to different threshold values. Please write your key observations based on F1 score for your model.	16
	(b)	Compare the performance of logistic regression, Decision Tree, Random Forest models on the given dataset. Analyze the results, clearly state the changes or optimizations you would make to each model before re-fitting, and justify your approach.	16
	(c)	Summarize the solution to the business problem as follows (8 marks): ● Feature Importance (4 marks): Identify and explain which features are most influential in solving the problem and why. ● Evaluation Metrics Performance (2 marks)Provide an analysis of the model's performance based on key evaluation metrics. ● Overall Results and Observations (2 marks): Summarize the overall findings, including key insights and their implications for the business problem.	8